Name: Ekta Narayan Sawant

Roll No: 2401103

Batch: T22

ASD&DO Experiment 01

Aim: Version control operations using Git.

This experiment maps to LO2

LO2: Use Git effectively for version control, source code management and collaborative team development.

Theory:

Command	Description (Short)
git status	Shows the current status of the working directory and staging area.
git init	Initializes a new Git repository or reinitializes an existing one.
git add <file>	Stages a specific file for the next commit.
git commit -m "message"	Creates a new commit with the staged changes.
git branch -M main	Renames the current branch to main.
git remote add origin <URL>	Adds a remote repository named origin.
git push -u origin main	Pushes the main branch and sets the upstream branch.
git branch	Lists all local branches.
git branch <branchname>	Creates a new branch.
git checkout <branchname>	Switches to the specified branch.
echo "text" > file.py	Creates a new file and writes content into it.
git add experimental.py	Stages experimental.py for commit.

git commit -m "first commit"	Commits the staged experimental file.
git push origin newbranch	Pushes the newbranch branch to the remote repository.
git checkout main	Switches to the main branch.
git shortlog	Displays a summary of commits grouped by author.
git log	Shows the complete commit history.
git log --oneline	Displays a compact, one-line commit history.
git diff	Shows unstaged changes between the working directory and the last commit.
git fetch origin	Downloads updates from the remote repository without merging.
git branch -a	Lists both local and remote branches.
git pull origin main	Fetches and merges changes from the remote main branch.
git remote -v	Displays the remote repository URLs.
gh auth status	Shows the current GitHub CLI authentication status.
gh auth login --scopes "repo"	Logs into GitHub CLI with the specified repository permissions.
git reset --hard HEAD~1	Removes the latest commit and discards all associated changes.

Output:


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git --version
git version 2.53.0
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ touch experiment.py
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ echo "Hello Git" > experiment
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.
Untracked files:
(use "git add <file>..." to include in what will be committed)
experiment.py
nothing added to commit but untracked files present (use "git add" to track)
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git add experimant.py
fatal: pathspec 'experimant.py' did not match any files
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git add experiment.py
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git commit -m "First commit"
[main 9afce9c] First commit
1 file changed, 1 insertion(+)
create mode 100644 experiment.py
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 297 bytes | 297.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/EktaSawant46/AI-ML-Internship
d5b6044..9afce9c main -> main
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git branch newbranch

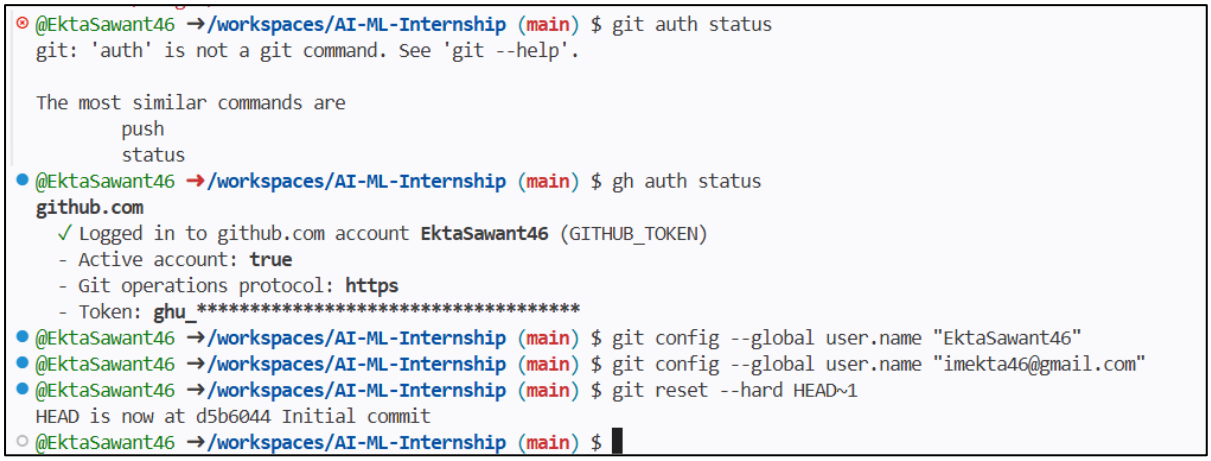
```
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git push
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git branch newbranch
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git branch
* main
  newbranch
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git checkout newbranch
Switched to branch 'newbranch'
@EktaSawant46 →/workspaces/AI-ML-Internship (newbranch) $ touch file1.py
@EktaSawant46 →/workspaces/AI-ML-Internship (newbranch) $ echo "newbranch file1" > file1.py
@EktaSawant46 →/workspaces/AI-ML-Internship (newbranch) $ git add file1.py
@EktaSawant46 →/workspaces/AI-ML-Internship (newbranch) $ git commit -m "commit to file1 "
[newbranch 3a3f9e7] commit to file1
 1 file changed, 1 insertion(+)
 create mode 100644 file1.py
@EktaSawant46 →/workspaces/AI-ML-Internship (newbranch) $ git push origin newbranch
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 333 bytes | 166.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'newbranch' on GitHub by visiting:
remote:      https://github.com/EktaSawant46/AI-ML-Internship/pull/new/newbranch
remote:
To https://github.com/EktaSawant46/AI-ML-Internship
 * [new branch]      newbranch -> newbranch
@EktaSawant46 →/workspaces/AI-ML-Internship (newbranch) $ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git log
```

```
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git log
commit 9afce9ca8933d10a9e63f6e2a99ece38991c6739 (HEAD -> main, origin/main, origin/HEAD)
Author: Ekta Narayan Sawant <imekta46@gmail.com>
Date:   Wed Jul 29 17:51:29 2026 +0000

    First commit

commit d5b604428639d54055926d470b819279944d3e60
Author: Ekta Narayan Sawant <imekta46@gmail.com>
Date:   Wed Jul 29 22:51:15 2026 +0530

    Initial commit
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git log --oneline
9afce9c (HEAD -> main, origin/main, origin/HEAD) First commit
d5b6044 Initial commit
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git diff
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git remote -v
origin  https://github.com/EktaSawant46/AI-ML-Internship (fetch)
origin  https://github.com/EktaSawant46/AI-ML-Internship (push)
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git fetch origin
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git pull origin main
From https://github.com/EktaSawant46/AI-ML-Internship
 * branch            main       -> FETCH_HEAD
Already up to date.
@EktaSawant46 →/workspaces/AI-ML-Internship (main) $ git branch -a
* main
  newbranch
  remotes/origin/HEAD -> origin/main
  remotes/origin/main
  remotes/origin/newbranch
```



Conclusion:

This experiment provided hands-on experience with the fundamental Git and GitHub workflow. It covered initializing a repository, checking repository status, staging and committing changes, creating and switching branches, managing remote repositories, pushing and pulling changes, viewing commit history, comparing changes, restoring files, amending commits, resetting commits, and authenticating with GitHub CLI. Overall, it helped build a strong understanding of version control, branch management, and collaboration using Git and GitHub.